

Simultaneous Design of FIR Filter Banks and Spatial Patterns for EEG Signal Classification

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Abstract—The spatial weights for electrodes called common spatial pattern (CSP) are known to be effective in EEG signal classification for motor imagery-based brain-computer interface (MI-BCI). To achieve accurate classification in CSP, it is necessary to find frequency bands that relate to brain activities associated with BCI tasks. Several methods that determine such a set of frequency bands have been proposed. However, the existing methods cannot find the multiple frequency bands by using only learning data. To address this problem, we propose discriminative filter bank CSP (DFBCSP) that designs finite impulse response filters and the associated spatial weights by optimizing an objective function which is a natural extension of that of CSP. The optimization is conducted by sequentially and alternatively solving subproblems into which the original problem is divided. By experiments, it is shown that DFBCSP can effectively extract discriminative features for MI-BCI. Moreover, experimental results exhibit that DFBCSP can detect and extract the bands related to brain activities of motor imagery.

Index Terms—Brain-computer interface (BCI), common spatial pattern (CSP), feature extraction, filter bank.

I. INTRODUCTION

BRAIN-COMPUTER interface (BCI) is a challenging application of signal processing and neuroscience [1]. BCIs capture brain activities associated with mental tasks and external stimuli and realize nonmuscular communication and control channel for conveying messages and commands to the external world [1]–[3]. Basically, noninvasive measurement devices such as EEG, magnetoencephalogram, and functional magnetic response imaging are widely used to observe the brain activities. Among them, because of its simplicity and low cost, EEG is a practical measurement device for use in engineering applications.

A promising realization of BCI is the so-called motor imagery-based BCI (MI-BCI) [2], [3]. Mu rhythm, which dis-

appears around motor cortex when a body moves, is one of features used in MI-BCI. It is also known that the imaginary movements of hands and feet evoke the change of mu rhythm in different brain regions [2], [3]. Therefore, by accurately capturing these changes from measured EEG signals in the presence of measurement noise and spontaneous components related to other brain activities, we can classify the EEG signal associated with imagination of different motor action such as right hand, left hand, or foot movement.

A well-known approach to extract the brain activity for MI-BCI is the so-called common spatial pattern (CSP) [1], [4], [5]. CSP gives a coefficient corresponding to each electrode in a multichannel EEG measurement system. The coefficients, which are the spatial weights, are determined from learning data in such a way that the variances of the signal extracted by the spatial weights differ between two classes (e.g., left- and right-hand movement imageries). Although together with the spatial weights an input EEG signal is successfully classified for the effective implementation, the observed signal should be bandpass-filtered to extract frequency components associated with motor imagery activities [4]. For the bandpass filtering, a passband of 7–30 Hz is widely chosen in MI-BCI, because mu and beta rhythms which are associated with motor imagery, are included in this band [2]. However, the optimum frequency band for classification is highly dependent on users and measurement environments. Therefore, for accurate classification, it is necessary to find the best frequency band or the associated filter parameters for each user and/or dataset. To this end, it would be simple to apply cross validation (CV) for searching the filter that gives the best classification accuracy for test data among several bandpass filters with different passbands. Hence, performance in classification depends on the choice of candidate filters, the selected filter is not always “optimal.” Moreover, a large size of the candidate set leads to higher computational cost. Therefore, it is crucial to establish a method of finding the optimal filter for classification in MI-BCI.

Recently, several approaches to this problem have been proposed [6]–[9]. Common spatio-spectral pattern (CSSP) [6] and common sparse spectral spatial pattern (CSSSP) [7] are the methods that obtain coefficients of a finite impulse response (FIR) filter by applying CSP to the combination of observed signals with the time-delayed signals. However, CSSP provides very poor frequency selectivity due to the limitation of a single delay. Unlike CSSP that provides different spectral patterns for each channel, CSSSP can provide a common spectral pattern for all channels. However, CSSSP needs the computationally expensive optimization because the optimization problem for the filter involves a sparsity cost and an extensive parameter

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tuning is needed. Spectrally weighted CSP (SPEC-CSP) [8] and iterative spatio-spectral patterns learning (ISSPL) [9] use iterative procedures for optimizing spatial weights and a filter, where the spatial weights are optimized by CSP and the filter parameterized by weights for the spectrum is found by an optimization problem based on other criteria. These two optimization problems are alternately solved; however, this alternating iteration is not guaranteed to be converged, because the cost functions for these two problems are different.

Recently proposed subband CSP (SBCSP) [10] and filter bank CSP (FBCSP) [11] are based on subband decomposition with multiple filters that have different passbands. As reported in BCI competition III [12] and IV, FBCSP achieves a high classification accuracy. In these methods, EEG signals are decomposed into multiple frequency components and CSP is separately applied for each frequency component. By using multiple filters such as these methods, features associated with brain activities such as mu and beta rhythms can be separately extracted. The result of FBCSP suggests that feature extraction using subband decomposition is an effective method to increase classification accuracy of MI-BCI. The methods using frequency decomposition need selection of feature values to achieve accurate classification. In order to select the feature values, the methods based on mutual information have been proposed [11], [13]. The other methods for selecting efficient filters are also proposed in [14] and [15]. In these methods, the filters in the filter bank have to be designed in advance and the fixed filter bank is independent of dataset. Although the parameters for the filters such as the passbands that accurately extract the feature components can change by users and measurement environments, the method using the fixed filter bank cannot use the filters optimized for a specific user.

In this paper, we address the problem of designing filters that extract multiple frequency components in spatial pattern-based EEG classification. We find these filters by maximizing the proposed cost function that is a natural extension of that of CSP. The maximization simultaneously provides the spatial weights and FIR filters. Unlike CSSP and CSSSP, the proposed method enables us to design multiple filters and the associated spatial patterns. This means that unlike SBCSP and FBCSP that divide the signal into subbands with a fixed filter bank and choose some of the subbands to form features, the proposed method directly designs the user-specific parameters for the filters such as the passbands that lead to the extraction of the features. This characteristic of the proposed method can solve the feature extraction problem extracting several components that have different frequency bands and different spatial patterns. For example, we can see feature extraction for MI-BCI as such feature extraction problem, because the components called mu and beta rhythms that have different frequency bands and different spatial patterns are known as the feature components for MI-BCI [1], [3]. We call the proposed method discriminative filter bank CSP (DFBCSP).

In order to introduce DFBCSP, we first define feature values to be extracted from an EEG signal by an FIR filter and spatial weights. We next formulate an optimization problem to find the optimal filters and the spatial weights. The optimization problem

can be separated to some subproblems and we show the solution for them that uses sequential and alternating optimization. Unlike SPEC-CSP and ISSPL, because DFBCSP optimizes the spatial weights and the frequency filters by the single criteria, it is guaranteed that the cost function monotonically increases by iteration in the alternating optimization procedure. Moreover, each subproblem is reduced to a generalized eigenvalue problem which can be solved by a well-established optimization method.

The rest of this paper is organized as follows. Section II reviews the CSP method and the CSP-based filter design/select methods (CSSP, CSSSP, SPEC-CSP, and FBCSP). Next, we propose DFBCSP in Section III. Section IV presents experimental results of classification of EEG signals during motor imageries to show the effectiveness of DFBCSP. We experimentally analyze DFBCSP by using artificial signals in Section V. Finally, the conclusion of this paper is presented in Section VI.

A. Notations

Throughout the paper, let $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{Z}$ denote the sets of all real numbers, all complex numbers, and all integers, respectively. A scalar is denoted by an italic letter, e.g., x or X . A vector is denoted by an italic bold lower case letter, e.g., $\mathbf{x}$. The i th entry of the vector $\mathbf{x}$ is denoted by $[\mathbf{x}]_i$, where i is a positive integer. A matrix is denoted by an italic bold upper case letter, e.g., $\mathbf{X}$. The element at the i th row and the j th column of a matrix $\mathbf{X}$ is denoted by $[\mathbf{X}]_{i,j}$, where i and j are positive integers. Transposes for $\mathbf{x}$ and $\mathbf{X}$ are denoted by $\mathbf{x}^T$ and $\mathbf{X}^T$, respectively. The absolute value of x is denoted by $|x|$. $\|\mathbf{x}\|_p$ is p -norm of a vector, $\mathbf{x} \in \mathbb{C}^N$, defined as $\|\mathbf{x}\|_p = (\sum_{i=1}^N |[x]_i|^p)^{1/p}$, and $\|\mathbf{x}\| = \|\mathbf{x}\|_2$. $\text{Span}\{\cdots\}$ represents a subspace spanned by vectors. The operator denoted by $\oplus$ gives the direct sum of two subspaces. The set having no elements is denoted by $\emptyset$. The orthogonal complement of a subspace is denoted by $\cdot^\perp$. The Kronecker delta δ_{ij} is defined as 1 for $i = j$ and 0 otherwise. The operator denoted by $a \bmod b$ takes a residue of dividing a by b . The operator denoted by $\Re$ takes a real part of a complex number. A Gaussian distribution with a mean m and a variance σ^2 is denoted by $\mathcal{N}(m, \sigma^2)$. A uniform distribution whose minimum and maximum values are denoted by a and b , respectively, is denoted by $\mathcal{U}(a, b)$.

II. CSP AND FILTER DESIGN/SELECTION METHODS

CSP is effectively used for the feature extraction and classification in a two-class MI-BCI. In this section, we review a basic CSP [4], [5] and its variants [6]–[8], [11].

A. CSP

Let $\mathbf{X} \in \mathbb{R}^{M \times N}$ be an observed signal, where M is the number of channels and N is the number of samples. CSP finds a spatial weight vector, $\mathbf{w} \in \mathbb{R}^M$, in such a way that the variance of a signal extracted by linear combination of $\mathbf{X}$ is minimized in a class [4], [5]. In BCI application, we do not directly use $\mathbf{X}$, but use the filtered signal described as $\hat{\mathbf{X}} = \mathcal{H}(\mathbf{X})$ in CSP, where $\mathcal{H}$ is a bandpass filter which passes the frequency

components related to brain activity of motor imagery. Denote the components of $\hat{\mathbf{X}}$ by $\hat{\mathbf{X}} = [\hat{\mathbf{x}}_1, \dots, \hat{\mathbf{x}}_N]$, where $\hat{\mathbf{x}}_n \in \mathbb{R}^M$ and n is the time index. The time average of the observed signal is given by $\boldsymbol{\mu} = (1/N) \sum_{n=1}^N \hat{\mathbf{x}}_n$. Then, the time variance of the extracted signal of $\hat{\mathbf{X}}$ is given by

$$\sigma^2(\mathbf{X}, \mathbf{w}) = \frac{1}{N} \sum_{n=1}^N |\mathbf{w}^T (\hat{\mathbf{x}}_n - \boldsymbol{\mu})|^2. \quad (1)$$

We assume sets of the learning data, $\mathcal{C}_1$ and $\mathcal{C}_2$, where $\mathcal{C}_d$ contains the signals belonging to class d , $d \in \{1, 2\}$ is a class label, and $\mathcal{C}_1 \cap \mathcal{C}_2 = \emptyset$. CSP finds the weight vector that minimizes the intraclass variance in $\mathcal{C}_c$ under the normalization of samples, where c is a class label. More specifically, for c fixed, CSP finds $\mathbf{w}_c$ by solving the following optimization problem [4], [5]:

$$\begin{aligned} \min_{\mathbf{w}} \quad & E_{\mathbf{X} \in \mathcal{C}_c} [\sigma^2(\mathbf{X}, \mathbf{w})] \\ \text{subject to} \quad & \sum_{d=1,2} E_{\mathbf{X} \in \mathcal{C}_d} [\sigma^2(\mathbf{X}, \mathbf{w})] = 1 \end{aligned} \quad (2)$$

where $E_{\mathbf{X} \in \mathcal{C}_d} [\cdot]$ denotes the expectation over $\mathcal{C}_d$. Then, (2) can be rewritten as

$$\min_{\mathbf{w}} \quad \mathbf{w}^T \boldsymbol{\Sigma}_c \mathbf{w}, \quad \text{subject to} \quad \mathbf{w}^T (\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2) \mathbf{w} = 1 \quad (3)$$

where $\boldsymbol{\Sigma}_d$, $d = 1, 2$, are defined as $\boldsymbol{\Sigma}_d = E_{\mathbf{X} \in \mathcal{C}_d} [\frac{1}{N} \sum_{n=1}^N (\hat{\mathbf{x}}_n - \boldsymbol{\mu})(\hat{\mathbf{x}}_n - \boldsymbol{\mu})^T]$. The solution of (3) is given by the generalized eigenvector corresponding to the smallest generalized eigenvalue of the generalized eigenvalue problem described as

$$\boldsymbol{\Sigma}_c \mathbf{w} = \lambda (\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2) \mathbf{w}. \quad (4)$$

Although the solution of (3) is given by the eigenvector corresponding to the smallest eigenvalue in (4), we can use the other eigenvectors for classification [16]. The M eigenvectors can be obtained by solving (4) as $\hat{\mathbf{w}}_1, \dots, \hat{\mathbf{w}}_M$, where $\hat{\mathbf{w}}_i$ is the eigenvector corresponding to the i th smallest eigenvalue of (4). We assume that the $2r$ eigenvectors are used for classification of unlabeled data, $\mathbf{X}$. Then, we obtain the feature vector, $\mathbf{y} \in \mathbb{R}^{2r}$, from $\mathbf{X}$ defined as

$$\mathbf{y} = [\sigma^2(\mathbf{X}, \hat{\mathbf{w}}_1), \dots, \sigma^2(\mathbf{X}, \hat{\mathbf{w}}_r), \sigma^2(\mathbf{X}, \hat{\mathbf{w}}_{M-r+1}), \dots, \sigma^2(\mathbf{X}, \hat{\mathbf{w}}_M)]^T. \quad (5)$$

For classification, $\mathbf{y}$ is an input to a classifier such as linear discriminant analysis (LDA) [17].

B. CSSP

The CSSP method obtains a spatial weight vector and a temporal filter by applying the combination of observed signals with time-delayed signals to CSP [6]. Let $\mathbf{X} \in \mathbb{R}^{M \times N}$ be an observed signal with M channels and N samples, and $\mathbf{X}_1$ and $\mathbf{X}_2$ be the subsignals included in $\mathbf{X}$. The components of $\mathbf{X}_1$ and $\mathbf{X}_2$ are defined as $[\mathbf{X}_1]_{m,n} = [\mathbf{X}]_{m,n}$ and $[\mathbf{X}_2]_{m,n} = [\mathbf{X}]_{m,n+\tau}$, respectively, where $m = 1, \dots, M$, $n = 1, \dots, N - \tau$, τ is a delay sample. Then, the τ -delay embedded signal is defined as $\mathbf{X}^\tau = [\mathbf{X}_1^T \mathbf{X}_2^T]^T \in \mathbb{R}^{2M \times (N-\tau)}$. $\mathbf{X}^\tau$ is used as an input signal of CSP to seek the spatial weight vector.

C. CSSSP

The CSSSP [7] method obtains a weight vector and an FIR filter by using delayed signals like CSSP. The main difference between both methods is that CSSSP achieves design of an FIR filter with order more than 2. When CSSSP designs an FIR filter with order T , the optimization problem of CSSSP has the optimization parameters consisting of M spatial weight coefficients and $T - 1$ coefficients for the FIR filter. Let $b_1, \dots, b_T$ be the coefficients of the FIR filter that we want to design. The optimization problem of CSSSP is

$$\begin{aligned} \max_{b_2, \dots, b_T} \max_{\mathbf{w}} \quad & \mathbf{w}^T \left(\sum_{\tau=0}^{T-1} \left(\sum_{j=1}^{T-\tau} b_j b_{j+\tau} \right) \boldsymbol{\Sigma}_c^\tau \right) \mathbf{w} + \frac{C}{T} \|\mathbf{b}\|_1 \\ \text{subject to} \quad & \mathbf{w}^T \left[\sum_{\tau=0}^{T-1} \left\{ \sum_{j=1}^{T-\tau} b_j b_{j+\tau} \right\} (\boldsymbol{\Sigma}_1^\tau + \boldsymbol{\Sigma}_2^\tau) \right] \mathbf{w} = 1 \end{aligned} \quad (6)$$

where b_1 is fixed to 1, $\mathbf{b}$ is a coefficient vector defined as $\mathbf{b} = [b_1, \dots, b_T]^T$, $\boldsymbol{\Sigma}_c^\tau$ is a correlation matrix between the signal $\mathbf{X}$ and the τ delayed signal $\mathbf{X}_\tau$, defined as $\boldsymbol{\Sigma}_c^\tau = E_{\mathbf{X} \in \mathcal{C}_c} [\mathbf{X} \mathbf{X}_\tau^T + \mathbf{X}_\tau \mathbf{X}^T]$ for $\tau > 0$ and $\boldsymbol{\Sigma}_c^0 = E_{\mathbf{X} \in \mathcal{C}_c} [\mathbf{X} \mathbf{X}^T]$, and C is a nonnegative regularization coefficient that affects sparsity of $\mathbf{b}$. Optimization techniques like gradient or line-search methods can be used for solving (6) if T is not too large [7].

D. SPEC-CSP

SPEC-CSP uses an iterative procedure to achieve optimization of both spatial weights and filters [8]. In SPEC-CSP, the feature value extracted from $\mathbf{X}$ is defined with the spatial weight vector $\mathbf{w}$ and the spectral weight, $\mathbf{B}$, as $v(\mathbf{X}, \mathbf{w}, \beta) = \mathbf{w}^T \mathbf{X} \mathbf{F}_N \mathbf{B} \mathbf{B}^T \mathbf{F}_N^T \mathbf{X}^T \mathbf{w} = \mathbf{w}^T (\sum_{k=1}^N b_k^2 \mathbf{V}_k) \mathbf{w}$ where $\mathbf{F}_N \in \mathbb{C}^{N \times N}$ is the discrete Fourier transform matrix defined as $[\mathbf{F}_N]_{k,l} = (1/\sqrt{N}) e^{-j2\pi(k-1)(l-1)/N}$, $k, l = 1, \dots, N$, $\mathbf{B}$ is a diagonal matrix defined as $\mathbf{B} = \text{diag}(b_1, \dots, b_N)$, $\mathbf{G} = \mathbf{X} \mathbf{F}_N = [\mathbf{g}_1, \dots, \mathbf{g}_N]$, $\mathbf{V}_k = \mathbf{g}_k \mathbf{g}_k^T$, and $\beta = [b_1^2, \dots, b_N^2]^T$. SPEC-CSP decides $2r$ weight vectors, $\hat{\mathbf{w}}_i$, and $2r$ filters, β_i , by alternately optimizing with CSP for $\hat{\mathbf{w}}_i$ and the optimization problem based on Fisher's criterion for β_i , where $i = 1, \dots, 2r$.

E. FBCSP

FBCSP selects some bandpass filters from a filter bank [11]. It comprises four stages: multiple bandpass filtering, spatial filtering by CSP, feature selection of the CSP features, and classification.

FBCSP first defines a bank of filters denoted by $\{\mathcal{H}_1, \dots, \mathcal{H}_{N_F}\}$. Let $\hat{\mathbf{X}}_j$ be the filtered signal by $\hat{\mathbf{X}}_j = \mathcal{H}_j(\mathbf{X})$. CSP is applied for each $\hat{\mathbf{X}}_j$ and spatial weights, $\mathbf{w}_j^i$, $i = 1, \dots, N_F$, $j = 1, \dots, M$, are found by solving eigenvalue problems of (4) as done for CSP, where $\mathbf{w}_j^i$ is the eigenvector corresponding to the j th largest eigenvalue in the eigenvalue problem in the filtered signals by $\mathcal{H}_i$. Features are selected out of a set of $\{\mathbf{w}_j^i | i = 1, \dots, N_F, j =$

$1, \dots, N_M, M - N_M + 1, \dots, M\}$ by a method based on mutual information between extracted features and its class labels, where N_M is the number of the spatial filter for a filter in the set. In [11], mutual information-based best individual feature (MIBIF) and mutual information-based naïve Bayesian Parzen window (NBPW) are introduced for selecting features. After selecting features, the selected r sets of indexes of the spatial weight and the filter are obtained as $\{\kappa_1, v_1\}, \dots, \{\kappa_r, v_r\}$, where $\kappa_i \in \{1, \dots, N_M, M - N_M + 1, \dots, M\}$ and $v_i \in \{1, \dots, N_F\}$. We denote the feature vector extracted from $\mathbf{X}$ by

$$\mathbf{y} = [\sigma^2(\hat{\mathbf{X}}_{v_1}, \mathbf{w}_{\kappa_1}^{v_1}), \dots, \sigma^2(\hat{\mathbf{X}}_{v_r}, \mathbf{w}_{\kappa_r}^{v_r})]^T. \quad (7)$$

III. DFBCSP

Because existing methods for designing filters based on CSP such as the CSSP, CSSSP, and SPEC-CSP methods seek for a single filter, the methods cannot find multiple frequency bands that relate to brain activities associated with BCI tasks. Although methods using subband decomposition such as FBCSP can select the bands [10], [11], [14], [15], these methods need designing the candidate filters in advance and do not achieve to directly design multiple filters.

In this section, DFBCSP is defined and its design procedure is illustrated. In DFBCSP, the observed multichannel EEG signal is temporally and spatially filtered by $\mathbf{h}_i$ and $\mathbf{w}_i$, respectively, and is transformed to a variance-based feature vector. Thus, the main problem behind this technique is finding $\mathbf{h}_i$ and $\mathbf{w}_i$ that efficiently extract EEG components related to the brain activity.

A. Feature Extraction

The feature extraction procedure is formulated in this section. The filtered signal of $\mathbf{X}$ denoted as $\hat{\mathbf{X}} = [\hat{\mathbf{x}}_1, \dots, \hat{\mathbf{x}}_K]$ can be defined as

$$\hat{\mathbf{x}}_n = \sum_{p=1}^P h_p \mathbf{x}_{n+P-p} \quad (8)$$

for $n = 1, \dots, K$, where P is a filter order of an FIR filter of which the coefficients are denoted by $h_1, \dots, h_P$ and $K = N - P + 1$. Moreover, the weighted-sum signal by the spatial weights, $\mathbf{w} = [w_1, \dots, w_M]^T$, is defined as

$$\tilde{\mathbf{x}}_n = \sum_{i=1}^M w_i [\hat{\mathbf{x}}_n]_i. \quad (9)$$

The procedure of temporally and spatially filtering for $\mathbf{X}$ is illustrated in Fig. 1.

The variance of $\tilde{\mathbf{x}}_n, n = 1, \dots, K$ that is used for feature values is

$$\alpha(\mathbf{X}, \mathbf{w}, \mathbf{h}) = \frac{1}{K} \sum_{n=1}^K \left| \tilde{\mathbf{x}}_n - \frac{1}{K} \sum_{i=1}^K \tilde{\mathbf{x}}_i \right|^2 \quad (10)$$

where $\mathbf{h}$ is the vector of the filter coefficients defined as $\mathbf{h} = [h_1, \dots, h_P]^T$.

The feature value of (10) can be transformed to matrix–vector form as follows. We define $\mathbf{A}_n, n = 1, \dots, K$, whose elements

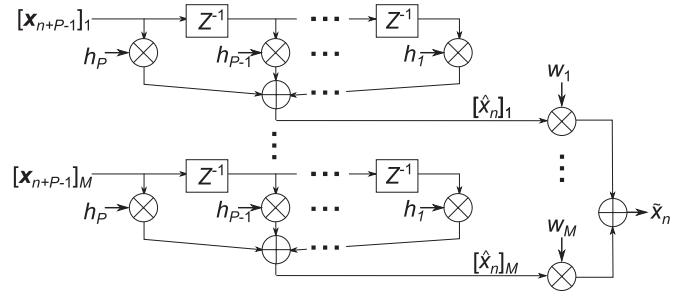


Fig. 1. Filtering of an observed signal by a temporal filter and spatial weights. In the figure, Z^{-n} is an operator of n samples delay, and $[\cdot]_i$ denotes an i th element of a vector.

are from $\mathbf{X}$ as

$$[\mathbf{A}_n]_{m,p} = [\mathbf{X}]_{m,n+P-p} \quad (11)$$

where $m = 1, \dots, M, p = 1, \dots, P$. Therefore, (10) can be modified to

$$\alpha(\mathbf{X}, \mathbf{w}, \mathbf{h}) = \frac{1}{K} \sum_{n=1}^K \left| \mathbf{w}^T \hat{\mathbf{A}}_n \mathbf{h} \right|^2 \quad (12)$$

where $\hat{\mathbf{A}}_n$ is defined as $\hat{\mathbf{A}}_n = \mathbf{A}_n - K^{-1} \sum_{m=1}^K \mathbf{A}_m$.

The feature vector, $\mathbf{y}$, using $\alpha(\mathbf{X}, \mathbf{w}, \mathbf{h})$ is defined as

$$\mathbf{y} = [\alpha(\mathbf{X}, \hat{\mathbf{w}}_1^{(1)}, \mathbf{h}_1), \dots, \alpha(\mathbf{X}, \hat{\mathbf{w}}_1^{(2r)}, \mathbf{h}_1), \dots, \alpha(\mathbf{X}, \hat{\mathbf{w}}_F^{(1)}, \mathbf{h}_F), \dots, \alpha(\mathbf{X}, \hat{\mathbf{w}}_F^{(2r)}, \mathbf{h}_F)]^T \quad (13)$$

where $\mathbf{h}_1, \dots, \mathbf{h}_F$ are the FIR filters and $\hat{\mathbf{w}}_i^{(m)}, m = 1, \dots, 2r$ is a m th spatial weight vector associated with $\mathbf{h}_i$.

B. Design Criterion and Optimization

For the feature value defined in (13), we design F frequency filters and the associated spatial weights. Let $\mathbf{w}_i$ and $\mathbf{h}_i$ for $i = 1, \dots, F$ be the spatial weights and the coefficients of filters, respectively. The underlying idea behind the proposed method is to find optimal value of both $\mathbf{w}_i$ and $\mathbf{h}_i$ by maximization of an expectation of $\alpha(\mathbf{X}, \mathbf{w}_i, \mathbf{h}_i)$ with respect to $\mathbf{X} \in \mathcal{C}_c$ samples under the normalization of an expectation of $\alpha(\mathbf{X}, \mathbf{w}_i, \mathbf{h}_i)$ over all of the observation. Additionally, we impose the orthonormality on $\mathbf{h}_i, i = 1, \dots, F$ to avoid the trivial solution. Therefore, we formulate the following maximization problem:

$$\begin{aligned} & \max_{\mathbf{w}_i, \mathbf{h}_i, i=1, \dots, F} \sum_{i=1}^F \hat{J}(\mathbf{w}_i, \mathbf{h}_i) \\ & \text{subject to} \quad \frac{\mathbf{h}_i^T \mathbf{h}_j}{\|\mathbf{h}_i\| \|\mathbf{h}_j\|} = \delta_{ij}, \quad i, j = 1, \dots, F \\ & \quad \mathbf{h}_k \in \mathcal{S}_k^\perp, \quad k = 1, \dots, F \end{aligned} \quad (14)$$

where

$$\hat{J}(\mathbf{w}, \mathbf{h}) = \frac{E_{\mathbf{X} \in \mathcal{C}_c} [\alpha(\mathbf{X}, \mathbf{w}, \mathbf{h})]}{\sum_{d=1,2} E_{\mathbf{X} \in \mathcal{C}_d} [\alpha(\mathbf{X}, \mathbf{w}, \mathbf{h})]} \quad (15)$$

$\mathcal{S}_i$ is any subspace in $\mathbb{R}^P$, and c is a class label chosen from 1 and 2.

It is difficult to simultaneously find all parameters. Therefore, we consider a subproblem for each filter index i . Then, we adopt a sequential optimization that finds the parameters one by one with respect to an filter index i . In the sequential optimization, we first find $\{\mathbf{w}_1, \mathbf{h}_1\}$, and then find $\{\mathbf{w}_2, \mathbf{h}_2\}$ under the constraint with $\{\mathbf{w}_1, \mathbf{h}_1\}$. By such sequential optimization, we can represent (14) with respect to each i as

$$\max_{\mathbf{w}_i, \mathbf{h}_i} \hat{J}(\mathbf{w}_i, \mathbf{h}_i), \quad \text{subject to } \mathbf{h}_i \in \hat{\mathcal{S}}_i^\perp \quad (16)$$

where $\hat{\mathcal{S}}_i$ is a subspace defined as $\hat{\mathcal{S}}_i = \text{Span}\{\mathbf{h}_1, \dots, \mathbf{h}_{i-1}\} \oplus \mathcal{S}_i \in \mathbb{R}^P$. Methods for choosing $\hat{\mathcal{S}}_i$ will be discussed in Section III-D. In (16), to optimize $\mathbf{w}_i$ and $\mathbf{h}_i$, we adopt an alternating optimization procedure based on alternating least square. Two subproblems that separately find $\mathbf{w}_i$ and $\mathbf{h}_i$ are obtained as follows.

The first subproblem is to optimize $\mathbf{w}_i$ while fixing $\mathbf{h}_i$. Define

$$\mathbf{R}_d(\mathbf{h}) = E_{\mathbf{X} \in \mathcal{C}_d} \left[\frac{1}{K} \sum_{n=1}^K \hat{\mathbf{A}}_n \mathbf{h} \mathbf{h}^T \hat{\mathbf{A}}_n^T \right] \quad (17)$$

for $d = 1, 2$. Then (16) can be written as

$$\max_{\mathbf{w}_i} \hat{J}_1(\mathbf{w}_i | \mathbf{h}_i) = \frac{\mathbf{w}_i^T \mathbf{R}_c(\mathbf{h}_i) \mathbf{w}_i}{\mathbf{w}_i^T (\mathbf{R}_1(\mathbf{h}_i) + \mathbf{R}_2(\mathbf{h}_i)) \mathbf{w}_i}. \quad (18)$$

The solution of (18) is given by the generalized eigenvector corresponding to the largest generalized eigenvalue of the generalized eigenvalue problem described as

$$\mathbf{R}_c(\mathbf{h}_i) \mathbf{w}_i = \lambda (\mathbf{R}_1(\mathbf{h}_i) + \mathbf{R}_2(\mathbf{h}_i)) \mathbf{w}_i. \quad (19)$$

We normalize $\mathbf{w}_i$ to unit norm.

The second subproblem is to optimize $\mathbf{h}_i$ while fixing $\mathbf{w}_i$. Define

$$\mathbf{Q}_d(\mathbf{w}) = E_{\mathbf{X} \in \mathcal{C}_d} \left[\frac{1}{K} \sum_{n=1}^K \hat{\mathbf{A}}_n^T \mathbf{w} \mathbf{w}^T \hat{\mathbf{A}}_n \right] \quad (20)$$

for $d = 1, 2$. Then, (16) can be written as

$$\begin{aligned} \max_{\mathbf{h}_i} \hat{J}_2(\mathbf{h}_i | \mathbf{w}_i) &= \frac{\mathbf{h}_i^T \mathbf{Q}_c(\mathbf{w}_i) \mathbf{h}_i}{\mathbf{h}_i^T (\mathbf{Q}_1(\mathbf{w}_i) + \mathbf{Q}_2(\mathbf{w}_i)) \mathbf{h}_i} \\ \text{subject to } \mathbf{h}_i^T \mathbf{v}_j^i &= 0, \quad j = 1, \dots, D_i \end{aligned} \quad (21)$$

where, $\mathbf{v}_j^i$, $j = 1, \dots, D_i$ are vectors that span $\hat{\mathcal{S}}_i$ as $\text{Span}\{\mathbf{v}_1^i, \dots, \mathbf{v}_{D_i}^i\} = \hat{\mathcal{S}}_i$. The solution to (21) is given by the following theorem.

Theorem 1: When matrices $\mathbf{Q}_c(\mathbf{w}_i)$ and $\mathbf{Q}_1(\mathbf{w}_i) + \mathbf{Q}_2(\mathbf{w}_i)$ are nonsingular, the solution of (21) is given by the unit-length generalized eigenvector corresponding to the largest generalized eigenvalue of the generalized eigenvalue problem described as

$$\mathbf{G} \mathbf{Q}_c(\mathbf{w}_i) \mathbf{h}_i = \zeta (\mathbf{Q}_1(\mathbf{w}_i) + \mathbf{Q}_2(\mathbf{w}_i)) \mathbf{h}_i \quad (22)$$

where

$$\begin{aligned} \mathbf{G} &= \mathbf{I}_P - \mathbf{V} (\mathbf{V}^T (\mathbf{Q}_1(\mathbf{w}_i) + \mathbf{Q}_2(\mathbf{w}_i))^{-1} \mathbf{V})^{-1} \\ &\quad \times \mathbf{V}^T (\mathbf{Q}_1(\mathbf{w}_i) + \mathbf{Q}_2(\mathbf{w}_i))^{-1} \end{aligned} \quad (23)$$

$\mathbf{V}$ is the matrix defined as

$$\mathbf{V} = [\mathbf{v}_1^i, \dots, \mathbf{v}_{D_i}^i] \in \mathbb{R}^{P \times D_i} \quad (24)$$

$\mathbf{I}_P$ is the $P \times P$ identity matrix, and ζ is an eigenvalue.

Proof: See Appendix A. ■

As mentioned previously, we alternately optimize $\mathbf{w}_i$ and $\mathbf{h}_i$ by solving optimization problem (18) and (21) for each index, i . We optimize the filters and the spatial weights in the order of $\{\mathbf{w}_1, \mathbf{h}_1\}$ to $\{\mathbf{w}_F, \mathbf{h}_F\}$. In the optimization, it is important to choose appropriate initial values for $\mathbf{h}_i$ and $\mathbf{w}_i$. However, we do not discuss the initialization problem in this paper. We adopt a simple initialization way as follows: initialize $\mathbf{h}_i$ as a random vector which is orthonormalized from $\mathbf{v}_1, \dots, \mathbf{v}_{D_i}$ by the Gram-Schmidt orthonormalization.

For the feature vector defined in (13), $\hat{\mathbf{w}}_i^{(m)}$, $i = 1, \dots, F$, $m = 1, \dots, 2r$ are decided as follows. By solving (19) with $\mathbf{h}_i$, we obtain MF spatial patterns as $\tilde{\mathbf{w}}_i^{(m)}$ for $i = 1, \dots, F$ and $m = 1, \dots, M$, where $\tilde{\mathbf{w}}_i^{(m)}$ is the unit-length eigenvector corresponding to the m th largest eigenvalue of (19). Then, $\hat{\mathbf{w}}_i^{(m)}$ are defined as $\hat{\mathbf{w}}_i^{(j)} = \tilde{\mathbf{w}}_i^{(j)}$, $\hat{\mathbf{w}}_i^{(r+j)} = \tilde{\mathbf{w}}_i^{(M-j+1)}$, for $i = 1, \dots, F$, $j = 1, \dots, r$.

C. Convergence of the Cost Function in Optimization

The alternating optimization of DFBCSP does not increase the cost functions and leads to a local maximum solution of (16). Here, we give the guarantee of convergence of the cost in the alternating iteration.

Proposition 1: The cost function of (16) monotonically increases or remains the same by iterations. That is

$$0 \leq \hat{J}(\mathbf{w}^{(k)}, \mathbf{h}^{(k)}) \leq \hat{J}(\mathbf{w}^{(k+1)}, \mathbf{h}^{(k+1)}) \leq 1 \quad (25)$$

where $\mathbf{w}^{(k)}$ and $\mathbf{h}^{(k)}$ are the parameters to be optimized at k th iteration.

Proof: See Appendix B. ■

D. Search Space for $\mathbf{h}_i$

The algorithm of the sequential update defined in (16) seeks $\mathbf{h}_i$ in the orthogonal complement of $\hat{\mathcal{S}}_i$. In this section, we introduce two ways to determine $\hat{\mathcal{S}}_i$.

1) *Subspace Spanned by Filter Coefficients Vectors:* This way uses as $\hat{\mathcal{S}}_i$ a subspace

$$\hat{\mathcal{S}}_i = \text{Span}\{\mathbf{h}_1, \dots, \mathbf{h}_{i-1}\}. \quad (26)$$

Because $\mathbf{h}_1, \dots, \mathbf{h}_{i-1}$ are orthogonal from each other, they are a set of an orthogonal basis of $\hat{\mathcal{S}}_i$.

2) *Subspace Spanned by Filter Coefficients Vectors and Its Shifted Vectors:* The subspace given as in (26) can yield similar bandpass filters in terms of the amplitude characteristics in the frequency domain. To avoid this redundancy, time-shifted filters of $\mathbf{h}_1, \dots, \mathbf{h}_{i-1}$ can be used for forming the subspace

$$\hat{\mathcal{S}}_i = \text{Span}\{\mathbf{h}_1, \dots, \mathbf{h}_{i-1}, \mathbf{h}'_1, \dots, \mathbf{h}'_{i-1}\} \quad (27)$$

where $\mathbf{h}'_j$, $j = 1, \dots, i-1$ are defined as $[\mathbf{h}'_j]_k = [\mathbf{h}_j]_{((k+1) \bmod P)}$, for $k = 1, \dots, P$.

This formulation is justified as follows. Taking into account that a filter vector can be decomposed into the superposition of sinusoids, we consider a simple filter given by a sinusoidal impulse response as $[\mathbf{h}_a]_n = \sin \omega_p(n-1)$, for $n = 1, \dots, P$. Suppose $\hat{\mathcal{S}}_b$ as the search space for $\mathbf{h}_b$ in the optimization problem of (16) and $\hat{\mathcal{S}}_b \ni \mathbf{h}_a$. The solution $\mathbf{h}_b$ can be a time-shifted filter of $\mathbf{h}_a$ such as $[\mathbf{h}_b]_n = \cos \omega_p(n-1)$, for $n = 1, \dots, P$, because of the orthogonality

$$\langle \mathbf{h}_a, \mathbf{h}_b \rangle = \sum_{n=0}^{N-1} \sin \omega_p n \cos \omega_p n = 0. \quad (28)$$

This implies that given a filter, the orthogonal complement to $\mathbf{h}_a$ can include a solution of filters that have the amplitude characteristics similar to the given filter.

The time-shifted solution such as $\mathbf{h}_b$ can be avoided by including $\cos \omega_p n$ or any time-shifted filter of $\mathbf{h}_a$ in $\mathcal{S}_a$, since $\cos \omega_p n = c_1 \sin \omega_p n + c_2 \sin \omega_p(n+m)$, $\forall n$, or $\mathbf{h}_b = c_1 \mathbf{h}_a + c_2 \mathbf{h}'_a$, where c_1 and c_2 are appropriate coefficients and $m \neq l\pi/\omega_p$, $m, l \in \mathbb{Z}$. Therefore, after obtaining $\mathbf{h}_a$, its shifted vector such as $\mathbf{h}_b$ is not in $\hat{\mathcal{S}}_{a+1}^\perp$ defined in (27).

By the method adding the shifted vectors to $\hat{\mathcal{S}}$, we can avoid obtaining some FIR filters which have similar amplitude characteristics. We will discuss the effects of this method in Section IV-B2.

E. Remarks

Because both DFBCSP and CSSSP design the FIR filter(s) by using the cost function of CSP, the resulting filters from DFBCSP and CSSSP can have similarity in frequency response, as seen in Fig. 10. Note that the main difference between both methods is that although by definition the CSSSP method finds a single FIR filter, the DFBCSP method finds plural filters that can extract plural components. Moreover, although the CSSSP method needs to approximate the correlations between a signal and its time shift and involves an $L1$ regularization, the proposed DFBCSP needs no approximation of the correlation and a simple alternative optimization.

IV. EXPERIMENT OF EEG SIGNAL CLASSIFICATION

We compare performance in classifying EEG signals during motor imagery using the proposed methods to that using major conventional methods (CSP, CSSP, CSSSP, SPEC-CSP, and FBCSP).

A. Data Description

We used dataset IVa from BCI competition III [12], which was provided by Fraunhofer FIRST (Intelligent Data Analysis Group) and Campus Benjamin Franklin of the Charité, University Medicine Berlin (Department of Neurology, Neurophysics Group) [18]. This dataset consists of EEG signals during right hand and right foot motor imageries. The EEG signals were recorded from five subjects labeled *aa*, *al*, *av*, *aw*, and *ay*. 118

EEG channels were measured at positions of the extended international 10/20-system. The measured signal was bandpass-filtered with the passband of 0.05–200 Hz, and then digitized at 1000 Hz with 16 bits ($0.1 \mu\text{V}$). During each experiment, visual cues told the subject which imagery task (left hand, right hand, or right foot) should be performed. The cue was indicated for 3.5 s and the subject performed the motor imagery for this period. The resting interval between two trials was randomized from 1.75 to 2.25 s. Only EEG trials for right hand and right foot were provided.

In this experiment, we furthermore applied to this data a Butterworth low-pass filter whose cutoff frequency is 50 Hz and the filter order is 4, and downsampled to 100 Hz. The dataset for each subject consisted of signals of 140 trials per class. The signal in each trial is extracted from the period of 0.5–3.5 s after a visual cue. Therefore, the length of the signal in each trial amounts to 3.0 s.

B. Result

First, we compare classification accuracy of the proposed method with those of conventional methods and we consider suitable parameters in DFBCSP. We discuss the amplitude characteristic of the filter and the spatial weights in each method in Section IV-B2. The behaviors of the cost functions and the optimization parameters in the alternating optimization in DFBCSP are illustrated in Section IV-B3.

1) *Classification Accuracy*: In the classification experiment, we extracted the feature vector from the EEG signal by each method, and classified the feature vector by an LDA classifier [17]. The feature vector was extracted as follows.

- 1) *CSP*: We applied the Butterworth bandpass filter with the passband of 7–30 Hz, and minimized the variance cost of the right-hand class in (2).
- 2) *CSSP*: We minimized the variance cost of the right-hand class in (2). The signals were applied a bandpass filter between 7–30 Hz as preprocessing.
- 3) *CSSSP1*: The order of the FIR filter was fixed to 16 [7]. The class label c in (6) was set to the right-hand class. The signals were applied a bandpass filter between 7–30 Hz as preprocessing.
- 4) *CSSSP2*: The classification procedure is same as CSSSP1 but different in the order of the filter. The order of the FIR filter was fixed to 40.
- 5) *SPEC-CSP*: The constants p' and q' were fixed to 0 and 1, respectively (for details of these constants, see [8]). The iteration number in alternating optimization was 30.
- 6) *FBCSP1*: The filter bank we designed in advance covers 4–40 Hz, which comprises 18 bandpass filters that cover a bandwidth of 2 Hz each. All filters were Chebyshev Type II filters. The number of spatial weights N_M in each band was set to 8. These parameters were decided by referring [11]. We selected the feature values used for classification by MIBIF with a naïve Bayesian Parzen window (NBPW) classifier [11].
- 7) *FBCSP2*: The classification procedure is same as FBCSP1 but different in the filter bank designed in advance. In

TABLE I
CLASSIFICATION ACCURACY [%] GIVEN BY 5×5 CV

Method	<i>aa</i>	<i>al</i>	Subject <i>av</i>	<i>aw</i>	<i>ay</i>	Ave.
CSP-Ref (S.D.) (Band [Hz]) (<i>r</i>)	90.36 ±3.29 10–15 1	98.36 ±1.35 12–24 10	76.64 ±5.56 10–13 2	98.64 ±0.73 10–16 2	94.50 ±2.92 10–11 2	91.70 ±2.83
CSP (S.D.) (<i>r</i>)	81.93 ±5.16 3	98.64 ±1.95 3	74.00 ±5.03 4	97.07 ±2.05 3	92.86 ±4.72 3	88.90 ±3.78
CSSP (S.D.) (<i>r</i> , <i>r</i>)	87.07 ±3.87 10, 2	95.71 ±2.73 6, 6	74.86 ±6.27 9, 6	98.07 ±1.99 7, 4	94.86 ±3.06 7, 4	90.11 ±3.58
CSSSP1 (S.D.) (<i>r</i> , <i>C</i>)	90.79 ±3.21 1, 0	99.11 ±1.47 5, 0.1	75.79 ±4.80 4, 1	99.57 ±0.78 4, 0.1	94.21 ±4.02 3, 0	91.89 ±2.86
CSSSP2 (S.D.) (<i>r</i> , <i>C</i>)	91.00 ±3.50 1, 0.1	98.93 ±1.46 1, 0	76.14 ±4.66 4, 1	99.07 ±1.47 1, 0	93.86 ±3.09 3, 5	91.80 ±2.84
SPEC-CSP (S.D.) (<i>r</i>)	83.93 ±4.49 2	99.21 ±1.47 1	72.50 ±5.64 5	98.93 ±1.46 2	92.36 ±5.17 3	89.30 ±3.65
FBCSP1 (S.D.) (<i>r</i>)	90.64 ±4.13 1	98.93 ±1.46 2	66.29 ±5.68 10	98.50 ±1.76 3	96.07 ±2.63 5	90.09 ±3.13
FBCSP2 (S.D.) (<i>r</i>)	89.29 ±4.08 2	99.07 ±1.47 3	69.79 ±5.00 5	98.71 ±1.50 8	95.64 ±2.42 4	90.50 ±2.88
FBCSP3 (S.D.) (<i>r</i>)	88.21 ±5.38 1	98.93 ±1.46 1	72.93 ±5.96 5	99.00 ±1.04 6	95.43 ±2.53 3	90.90 ±3.27
DFBCSP1 (S.D.) (<i>r</i> , <i>F</i>)	92.29 ±2.99 1, 3	99.29 ±1.36 9, 3	78.07 ±3.77 1, 2	99.29 ±0.89 1, 1	95.07 ±3.10 3, 6	92.80 ±2.43
DFBCSP2 (S.D.) (<i>r</i> , <i>F</i>)	91.00 ±3.84 1, 2	99.14 ±1.47 5, 2	76.43 ±4.16 1, 1	99.57 ±0.78 1, 1	95.64 ±3.14 4, 3	92.36 ±2.68

The figure with $\pm$ represents the standard deviation (SD). The values below the columns shown in the classification accuracies are the parameters for each subject.

FBCSP2, the filter bank covers 4–40 Hz, which comprises nine bandpass filters that cover a bandwidth of 4 Hz each.

- 8) *FBCSP3*: The classification procedure is also same as FBCSP1 but different in the filter bank designed in advance. In FBCSP3, the filter bank covers 4–40 Hz, which comprises six bandpass filters that cover a bandwidth of 6 Hz each.
- 9) *DFBCSP1*: The feature vector defined in (13) was used, where the class corresponding to the right-hand movement imagery was chosen as the class *c*. The filter order *P* was set to 40. In optimization, we stopped iteration when the error of the cost function between successive iterations becomes under 10^{-5} . We adopt (26) as $\hat{S}_i$.
- 10) *DFBCSP2*: The classification procedure is same as DFBCSP1 but we adopt (27) as $\hat{S}_i$ to avoid obtaining redundant filters.
- 11) *CSP-Ref*: We obtained classification accuracies by 5×5 CV using various passbands for the bandpass filter used for preprocessing in CSP. The passbands are represented as f_l – f_u Hz for $f_l = 1, \dots, 48$ and $f_u = f_l + 1, \dots, 49$. We show the best classification accuracy and the passband performing the best classification accuracy in each subject as CSP-Ref in Table I.

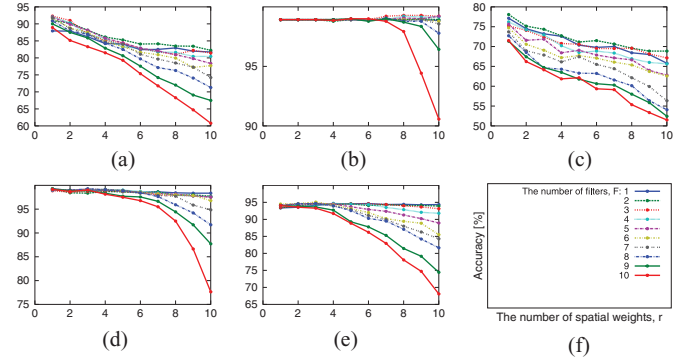


Fig. 2. Relations between classification accuracy and parameters in DFBCSP1. (a) Subject *aa*. (b) Subject *al*. (c) Subject *av*. (d) Subject *aw*. (e) Subject *ay*. (f) Axis labels and legends.

In the column below the column showing the classification accuracy in Table I, we show the parameters that we tuned to obtain the best classification accuracy by using 5×5 CV for each method and subject in advance. The parameters tested and decided in advance were as follows. The parameter for the dimension of the feature vector *r* was chosen out of $\{1, \dots, 20\}$. The number of filters *F* in DFBCSP was chosen out of $\{1, \dots, 10\}$. The delay τ in CSSP was chosen out of $\{1, \dots, 15\}$. The regularization parameter *C* in CSSSP was chosen out of $\{0, 0.01, 0.1, 0.2, 0.5, 1, 2, 5\}$. The passband in CSP-Ref is also tuned in advance. We calculated the logarithm of the feature values extracted by each method and the feature vectors consisted of these logarithm features. Finally, the feature vector extracted by each method was projected into the 1-D space determined by LDA and was classified by a threshold that is the middle point of two class averages over the learning samples. In Table I, both of the proposed methods outperform the conventional filter designing methods. Moreover, the proposed methods achieve similar accuracy of CSP-Ref that manually searches a filter performing the best classification accuracy by CV. On average over subjects, DFBCSP1 performs the smallest SD for CV.

Fig. 2 shows the relations between classification accuracy and parameters, the number of FIR filters [*F* in (13)], and the number of associated spatial weights [*r* in (13)], in DFBCSP1. Regardless of the number of filters, DFBCSP1 achieves the high accuracy when *r* = 1.

2) *Filters*: We show the amplitude characteristics of the filters for subject *aa* used/selected by each method shown in Section IV-B1 in Figs. 3 and 4. These filters and spatial weights were used/selected under optimization using all samples. In DFBCSP1, because it only has the constraint of orthogonality between coefficient vectors of the FIR filters, the filters can take similar amplitude characteristics. However, in the example shown in Fig. 4, we can observe that the filters extract three different frequency bands, around 12, 25, and 38 Hz. Because the spatial weights shown in Figs. 3 and 4 except Fig. 4(f) are similar, the results indicate that all methods extract the components observed in the same part of brain. The passbands shown in Figs. 3(e), (f), and 4(a) are similar to that

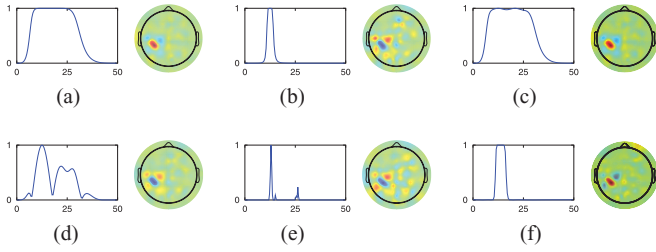


Fig. 3. Filters and the associated spatial weights in each method for subject *aa*. (a) CSP. (b) CSP-Ref. (c) CSSP. (d) CSSSP1. (e) SPEC-CSP. (f) FBCSP2 (first filter).

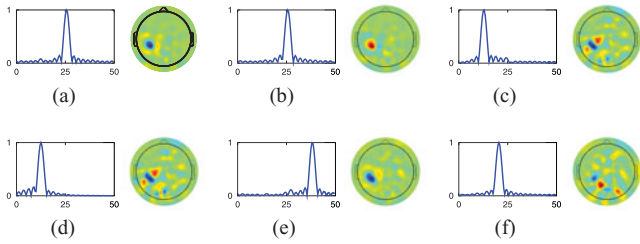


Fig. 4. Filters and the associated spatial weights designed by DFBCSP1 in subject *aa*. (a) First filter. (b) Second filter. (c) Third filter. (d) Fourth filter. (e) Fifth filter. (f) Sixth filter.

of CSP-Ref. This implies that these methods can find a passband that is effective for classification. In addition, comparing the passbands selected by FBCSP and designed by DFBCSP, we can observe that both methods specify similar passbands. Moreover, because the spatial weights in Fig. 4 are similar and the large coefficients of the spatial weights concentrate around left motor cortex, it is suggested that these passbands found by DFBCSP relate to brain activity of motor imagery. Besides, we can see in Fig. 4(f) the different spatial weight vector from those seen in Fig. 4(a)–(e). This spatial weight vector suggests the relation between the MI task and the components generated in the outside of the motor cortex.

Figs. 5 and 6 show the amplitude characteristics of the filters obtained by DFBCSP1 and DFBCSP2, respectively, for subject *ay*. We can observe that the amplitude characteristics shown in Fig. 5(a) and (c), and Fig. 5(d) and (e) are similar, respectively. On the other hand, all characteristics in Fig. 6 are different from each other. Additionally, as shown in Table I, the numbers of filters of DFBCSP2 for all subjects are fewer than those of DFBCSP1 and the maximum accuracies for DFBCSP1 and DFBCSP2 are almost similar. Therefore, incorporating time-shifted filter vectors into $\hat{S}_i$ is effective in avoiding giving redundant filters.

3) *Convergence of DFBCSP*: We illustrate the behavior of the cost function in the alternating optimization in Fig. 7. In the experiment, we found the optimization parameters in a similar way of DFBCSP1 shown in Section IV-B1 and used all data as the learning samples for the optimization. We can observe that the cost function monotonically increases and that the cost converges until ten iterations.

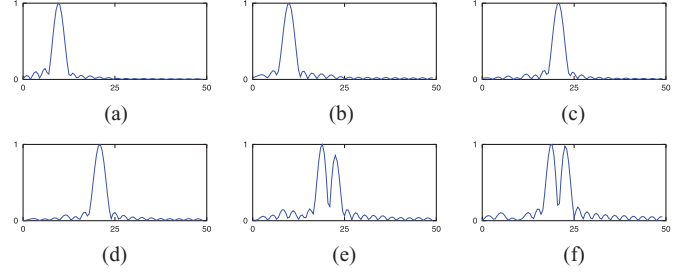


Fig. 5. Amplitude characteristics of the filters designed by DFBCSP1 with in subject *ay*. (a) First filter. (b) Second filter. (c) Third filter. (d) Fourth filter. (e) Fifth filter. (f) Sixth filter.

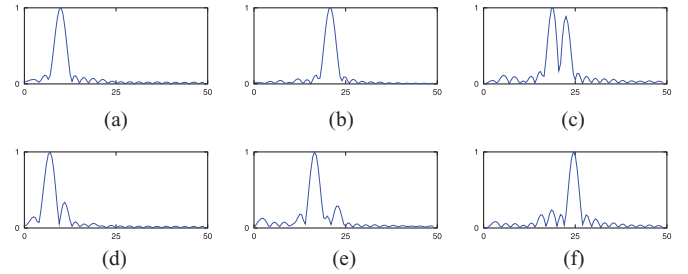


Fig. 6. Amplitude characteristics of the filters designed by DFBCSP2 with in subject *ay*. (a) First filter. (b) Second filter. (c) Third filter. (d) Fourth filter. (e) Fifth filter. (f) Sixth filter.

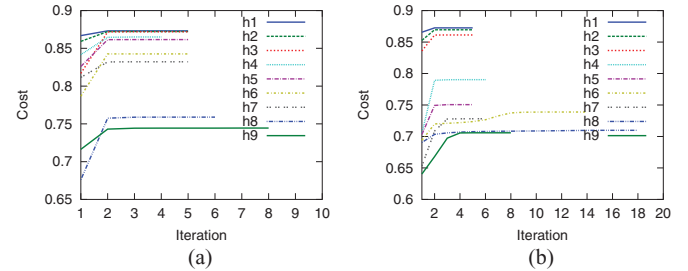


Fig. 7. Changes of the cost functions for iterations in the alternating optimization for the filters $h_1, \dots, h_9$. The dataset is of subject *aa*. (a) DFBCSP1. (b) DFBCSP2.

We show an example of the optimization process for DFBCSP1 fourth filter in Fig. 8. The initial value shown in Fig. 8(a) is a random vector that is orthogonal to the first, second, and third filters. We can see in Fig. 8(b) and (c) that optimization parameters are changed by updating other parameters. Additionally, we can observe from Figs. 7 and 8 that optimization parameters at the second iteration almost reach the results of final iteration.

V. EXPERIMENTAL ANALYSIS OF DFBCSP

We give an analysis of DFBCSP by a toy experiment with an artificial signal in this section. We assume a two-class BCI where observed EEG signals are modeled by a mixture of narrow-band signals (see Fig. 9). In this model, a trial signal belonging to class

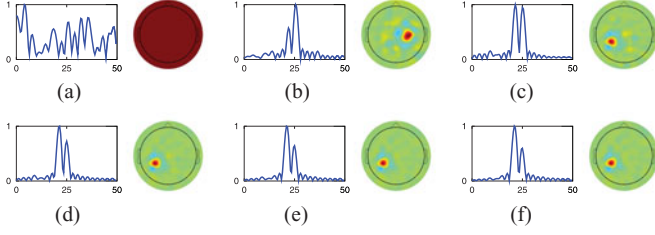


Fig. 8. Values of the optimization parameters in each iteration in the optimization process for the DFBCSP1 fourth filter in dataset of subject *aw*. For each figure, the left one is the amplitude characteristic of $\mathbf{h}_4$ and the right one is the topographically plotted spatial weights $\mathbf{w}_4$. (a) Initial value. (b) First iteration. (c) Second iteration. (d) Third iteration. (e) Fourth iteration. (f) Fifth iteration (final).

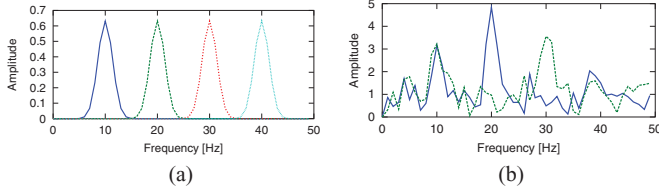


Fig. 9. Spectrum of the artificial signals. (a) Source signals. (b) Observed signals.

c is given by

$$\mathbf{x}[n] = \sum_{i=1}^{N_s} s_i[n] \mathbf{a}_i^{(c)} + \boldsymbol{\eta}, \quad n = 0, \dots, N-1 \quad (29)$$

where c is a class label taking either 1 or -1 , $\mathbf{x}[n] \in \mathbb{R}^M$ is a vector representing a signal at a discrete time point n observed, N is the number of time samples for a trial, M is the number of channels, $s_i[n] \in \mathbb{R}$ is an i th source signal of feature components, N_s is the number of the source signals, $\mathbf{a}_i^{(c)} \in \mathbb{R}^M$ is a vector defined by $\mathbf{a}_i^{(c)} = [a_{i1}^{(c)}, \dots, a_{iM}^{(c)}]^T$, $a_{im}^{(c)} \in \mathbb{R}$ is an amplitude of $s_i[n]$ in the m th channel for class c , and $\boldsymbol{\eta} \in \mathbb{R}^M$ is a stochastic noise. The source signals $s_i[n]$ are generated as $s_i[n] = \Re[\sum_{k=0}^{N-1} S_i[k] e^{j\theta} e^{j\frac{2\pi k}{N}n}]$, where $S_i[k] \in \mathbb{R}$ represents a discrete spectrum and $\theta \in \mathbb{R}$ is a stochastic phase of the source signals.

Specifically, the artificial signals we used in the experiment were generated as follows. The number of channel M was set to 5, the number of samples N was set to 100, the sampling frequency was assumed to be 100 Hz, the number of sources N_s was set to 4, and the number of trials was set to 200 (100 for each class). The spectra of source signals, $|S_1[k]|, \dots, |S_4[k]|$, are illustrated in Fig. 9(a) and the phase θ is uniformly distributed, $\mathcal{U}(0, 2\pi)$. The amplitudes, $\mathbf{a}_i^{(-1)}$ and $\mathbf{a}_i^{(1)}$ for $i = 1, \dots, 4$, are decided at random and normalized such that $\|\mathbf{a}_i^{(c)}\| = 1$. The noise $[\boldsymbol{\eta}]_m$ is Gaussian of $\mathcal{N}(0, 0.1)$. Examples of the spectra of observed signals generated by the aforementioned conditions are illustrated in Fig. 9(b).

We designed filters for the set of the artificial signals by CSSSP, SPEC-CSP, FBCSP-A, FBCSP-B, and DFBCSP2. The procedures are same as the procedures shown in Section IV-B.

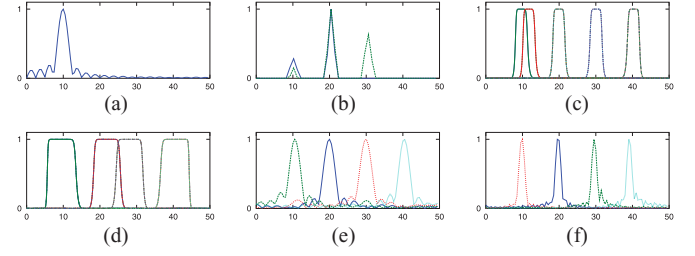


Fig. 10. Amplitude characteristics of the filters designed by each methods. The bands of the filters designed by DFBCSP correspond to the bands of source signals. (a) CSSSP. (b) SPEC-CSP. (c) FBCSP-A. (d) FBCSP-B. (e) DFBCSP2 ($P = 40$). (f) DFBCSP2 ($P = 80$).

In the procedures for designing the filters, the regularization parameter C and the order of the FIR filter T in CSSSP were set to 1 and 16, respectively. The filter bank used in FBCSP-A covers 7–42 Hz, which comprises 18 bandpass filters that cover a bandwidth of 2 Hz each. The filter bank used in FBCSP-B covers 7–49 Hz, which comprises seven bandpass filters that cover a bandwidth of 6 Hz each. The orders of the FIR filters in DFBCSP were set to 40 and 80, respectively.

Results of the simulation are summarized in Fig. 10, where we can see that the proposed DFBCSP works very well in accurately extracting narrow-band components. The filter designed by CSSSP only extracts the 10-Hz components and does not extract the other feature components as seen in Fig. 10(a). The filters designed by SPEC-CSP extract some components as illustrated in Fig. 10(d). However, more than two components are extracted by one filter and the filters do not separately extract each feature component as illustrated in Fig. 10(b). FBCSP and DFBCSP can separately extract each component by plural filters as seen in Fig. 10(c) and (d). In FBCSP, accuracy of estimation of the centers of the spectrum of the source signals depends on the filter bank. On the other hand, DFBCSP, which does not need the fixed filter bank, finds the centers accurately.

We also examined how the order of the filter P affects the results in Fig. 10(e) and (f). The bandwidths of each filter in Fig. 10(f) are narrower than those in Fig. 10(e). In DFBCSP, a larger filter order can achieve a narrower bandwidth of the filters.

VI. CONCLUSION

We have proposed a novel method for classification of EEG signals during motor imagery. Unlike conventional methods, the proposed DFBCSP algorithm can design multiple FIR filters and the associated spatial weights by maximizing the cost that is a natural extension of that of CSP. We have shown the optimization procedure for the problem of DFBCSP that conducted by sequentially and alternatively solving subproblems into which the original problem is divided. Moreover, we have shown that the subproblem can be solved by a generalized eigenvalue problem. By experiment, we have demonstrated that DFBCSP achieves a high classification accuracy. Our experimental results suggest also that DFBCSP can specify the frequency bands which associate with the brain activity of interest.

In the toy experiment using the artificial signal, we have shown an efficiency of DFBCSP for a feature extraction problem including plural feature components. We have also shown an effect of the filter order on filters designed by DFBCSP. Although the filter order can roughly control the bandwidths of the filters, seeking of suitable bandwidths for each filter should be solved.

APPENDIX A

PROOF OF THEOREM 1

For convenience, we define $\mathbf{Q}_c = \mathbf{Q}_c(\mathbf{w}_i)$, $\mathbf{Q} = \mathbf{Q}_1(\mathbf{w}_i) + \mathbf{Q}_2(\mathbf{w}_i)$, and $\mathbf{v}_j^i = \mathbf{v}_j$, $j = 1, \dots, D_i$. Any vector $\mathbf{h}_i$ can be normalized such that $\mathbf{h}_i^T \mathbf{Q} \mathbf{h}_i = 1$ and $\hat{J}_2(\mathbf{h}_i | \mathbf{w}_i)$ keeps unchanged. Therefore, the maximization of $\hat{J}_2(\mathbf{h}_i | \mathbf{w}_i)$ is equivalent to the maximization of $\mathbf{h}_i^T \mathbf{Q}_c \mathbf{h}_i$ with the constraint that $\mathbf{h}_i^T \mathbf{Q} \mathbf{h}_i = 1$. Then, the Lagrangian of (21) is

$$L = \mathbf{h}_i^T \mathbf{Q}_c \mathbf{h}_i - \zeta(\mathbf{h}_i^T \mathbf{Q} \mathbf{h}_i - 1) - \sum_{j=1}^{D_i} \nu_j \mathbf{h}_i^T \mathbf{v}_j \quad (30)$$

where ζ and $\nu_1, \dots, \nu_{D_i}$ are Lagrange multipliers. The partial derivative of L with respect to $\mathbf{h}_i$ is

$$\frac{\partial L}{\partial \mathbf{h}_i} = 2\mathbf{Q}_c \mathbf{h}_i - 2\zeta \mathbf{Q} \mathbf{h}_i - \sum_{j=1}^{D_i} \nu_j \mathbf{v}_j. \quad (31)$$

Then, $\partial L / \partial \mathbf{h}_i$ is zero when $\zeta = (\mathbf{h}_i^T \mathbf{Q} \mathbf{h}_i) / (\mathbf{h}_i^T \mathbf{Q}_c \mathbf{h}_i)$, because by left-multiplying $\mathbf{h}_i^T$ in $\partial L / \partial \mathbf{h}_i = 0$, we obtain

$$2\mathbf{h}_i^T \mathbf{Q} \mathbf{h}_i - 2\zeta \mathbf{h}_i^T \mathbf{Q}_c \mathbf{h}_i - \sum_{j=1}^{D_i} \nu_j \mathbf{h}_i^T \mathbf{v}_j = 0 \quad (32)$$

where the third term in the left-hand side is zero because $\mathbf{h}_i^T \mathbf{v}_j = 0$ for $i \neq j$. We define $\mathbf{V}$ by (24) and $\boldsymbol{\nu}$ as $\boldsymbol{\nu} = [\nu_1, \nu_2, \dots, \nu_{D_i}]^T$. Then, multiplying $\partial L / \partial \mathbf{h}_i = 0$ by $\mathbf{V}^T \mathbf{Q}^{-1}$ is $2\mathbf{V}^T \mathbf{Q}^{-1} \mathbf{Q}_c \mathbf{h}_i - \mathbf{V}^T \mathbf{Q}^{-1} \mathbf{V} \boldsymbol{\nu} = 0$, because $2\zeta \mathbf{V}^T \mathbf{Q}^{-1} \mathbf{Q} \mathbf{h}_i = 0$. Thus

$$\boldsymbol{\nu} = 2(\mathbf{V}^T \mathbf{Q}^{-1} \mathbf{V})^{-1} \mathbf{V}^T \mathbf{Q}^{-1} \mathbf{Q}_c \mathbf{h}_i. \quad (33)$$

By substituting (33) into $\partial L / \partial \mathbf{h}_i = 0$, we obtain

$$\begin{aligned} 2\mathbf{Q}_c \mathbf{h}_i - 2\zeta \mathbf{Q}_c \mathbf{h}_i - 2\mathbf{V}(\mathbf{V}^T \mathbf{Q}^{-1} \mathbf{V})^{-1} \mathbf{V}^T \mathbf{Q}^{-1} \mathbf{Q}_c \mathbf{h}_i &= 0 \\ \mathbf{G} \mathbf{Q}_c \mathbf{h}_i &= \zeta \mathbf{Q} \mathbf{h}_i \end{aligned} \quad (34)$$

where $\mathbf{G}$ is defined in (23). Since ζ is the criterion to be maximized, the maximum solution of (21) is achieved by the unit-length generalized eigenvector corresponding to the largest generalized eigenvalue of (22).

APPENDIX B

PROOF OF PROPOSITION 1

We first consider an upper bound of $\hat{J}(\mathbf{w}, \mathbf{h})$. For convenience, we define $P_d = E_{\mathbf{X} \in \mathcal{C}_d}[\alpha(\mathbf{X}, \mathbf{w}, \mathbf{h})]$ for $d = 1, 2$. Therefore, we can represent $\hat{J}(\mathbf{w}, \mathbf{h})$ as $\hat{J}(\mathbf{w}, \mathbf{h}) = P_c / (P_1 + P_2)$. Then, $\hat{J}(\mathbf{w}, \mathbf{h}) = P_c / (P_1 + P_2) \leq 1$, because, $P_1, P_2 < P_1 + P_2$. It is the upper bound of the cost function, $\hat{J}(\mathbf{w}, \mathbf{h})$.

Note that $\hat{J}(\mathbf{w}^{(k)}, \mathbf{h}^{(k)}) = \hat{J}_1(\mathbf{w}^{(k)} | \mathbf{h}^{(k)}) = \hat{J}_2(\mathbf{h}^{(k)} | \mathbf{w}^{(k)}) > 0$. Given $\mathbf{w}^{(k+1)}$ by maximizing (18) with $\mathbf{R}_d(\mathbf{h}^{(k)})$. Then, the relationship, $0 \leq \hat{J}_1(\mathbf{w}^{(k)} | \mathbf{h}^{(k)}) \leq \hat{J}_1(\mathbf{w}^{(k+1)} | \mathbf{h}^{(k)}) \leq 1$, is given, because $\hat{J}_1(\mathbf{w} | \mathbf{h}^{(k)})$ with the constraint and fixing $\mathbf{h}^{(k)}$ achieves its maximum by $\mathbf{w}^{(k+1)}$. Moreover, the relationship yields $0 \leq \hat{J}(\mathbf{w}^{(k)}, \mathbf{h}^{(k)}) \leq \hat{J}(\mathbf{w}^{(k+1)}, \mathbf{h}^{(k)}) \leq 1$. In a similar way, it holds that $0 \leq \hat{J}(\mathbf{w}^{(k)}, \mathbf{h}^{(k)}) \leq \hat{J}(\mathbf{w}^{(k)}, \mathbf{h}^{(k+1)}) \leq 1$. This completes the proof.

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